

Chapter 10

Anti-Aging Medicine as a Game Changer for Long-Lasting Space Missions



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Abstract Many studies have highlighted that astronauts involved in long-lasting space missions are at risk of developing pathologies similar to those caused by aging. This finding has led several scholars to raise ethical concerns about plans for exploration and colonization of other celestial bodies. This research, while recognizing the hitherto expressed medical and ethical concerns as founded, highlights the fact that human enhancement, and in particular anti-aging therapies, could be the game changer. As it will be showed, anti-aging medicine is progressing at an increasingly accelerated pace. Quantitative and qualitative methods are implemented to provide evidence of this fact. First, the author presents the results of a scientometric analysis of trends in the field of anti-aging medicine. The available data clearly show that studies in this field grow at an almost exponential rate. Then, the author resorts to qualitative meta-analysis to review several medical innovations such as stem cell transplants, organ regeneration, genetic editing, heterochronic parabiosis, telomere lengthening, growth hormone treatments, and some drug therapies, starting from those based on lithium. Given the amount and quality of studies in the field of regenerative medicine, it seems justified to cultivate a moderate optimism regarding future long-lasting space missions. If safe remedies to rejuvenate the human organism in its entirety are discovered, ethical doubts about space exploration and colonization should *ipso facto* dissolve.

10.1 Background

A review of the most recent literature on the medical aspects of space exploration is puzzling. On the one hand, studies are pointing out that long-term space missions may have anti-aging effects. For instance, a team of researchers observed that “in *Caenorhabditis elegans*, spaceflight suppressed the formation of transgenically expressed polyglutamine aggregates, which normally accumulate with increasing

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age.” The researchers also observed that “the inactivation of each of seven genes that were down-regulated in space extended lifespan on the ground” (Honda et al. 2012).

Experiments on nematodes showing that aging slows down in space prompted investigations of spaceflight’s anti-aging effects in humans. As is well known, the atmosphere and the geomagnetic field protect life from solar wind and cosmic rays. By analyzing astronauts’ heart rate variability (HRV), Otsuka et al. (2019) concluded that “the magnetic field can affect and enhance HRV indices involved in longevity, notably during the daytime and evening, probably in association with an elevated activity of brain’s DMN [Default Mode Network, A/N] with help from the circadian clock.” The researchers observed that anti-aging HRV indices “were significantly increased on days of higher magnetic activity”.

On the other hand, there is copious scientific literature pointing out the deleterious effects of spaceflights on human health. Strollo et al. (2018) state that “astronauts coming back from long-term space missions present with different health problems potentially affecting mission performance, involving all functional systems and organs and closely resembling those found in the elderly”. In other words, it seems that space missions have aging effects, rather than anti-aging ones. These apparently aging effects concern several organs, tissues, and systems, such as bones, immune system, blood, cartilage, and eyes. More in detail, Cappellesso et al. (2015) underline that “osteoporosis is one of the established major consequences of long-duration spaceflights in astronauts seriously undermining their health after their returning on Earth” and specify that “astronauts typically lose more bone mass during one month than postmenopausal women on Earth lose in one year”.

That same fact was noticed by Cavanagh et al. back in 2005. These researchers showed that “bone loss in the lower extremities and lumbar spine is an established consequence of long-duration human space flight” and suggested possible countermeasures for the problem. It has been suggested that the class of drugs most commonly used to treat osteoporosis in postmenopausal women, that is anti-resorptives such as risedronate sodium, alendronate sodium, raloxifene, and zoledronic acid, could be also used to treat astronauts. Besides, the authors mentioned the possible use of supplements such as vitamin D and calcium, and of “anabolic agents such as parathyroid hormone (PTH) and teriparatide (rhPTH [1–34])”. The idea of dealing with the problem of bone loss induced by long-term spaceflights by way of genetic enhancement has also been explored, for instance by Szocik et al. (2019a).

Sonnenfeld et al. (2003) offer evidence that immune response is another of the many physiologic functions of mammals affected by exposure to space flight conditions and concluded that “such alterations could lead to compromised defenses against infections and tumors”. More recently, Crucian et al. (2018) have confirmed that the immune system is subject to dysregulation during spaceflight and suggested potential countermeasures for deep space exploration missions. The reactivation of latent herpesviruses in astronauts for the duration of a 6-month orbital spaceflight is a well-known phenomenon. It has been observed that spaceflight dysregulates adaptive immunity, heightens innate immunity, and alters the interaction between adaptive

and innate immunity. As a consequence, it may happen that astronauts experience persistent hypersensitivity reactions during flight.

Red blood cells are also negatively affected by the condition of space flights. De Santo et al. (2005) point out that “a state of anemia associated with reduced erythropoietin levels and reduced plasma volume” was already observed after the very early crewed missions in space, and that “the reduction in red blood cell mass is driven by a process of selective hemolysis, which has been named neocytolysis”.

In this case also, a parallel could be traced between the clinical condition of the astronauts and that of the elderly. It is well known that anemia is very common in seniors. Ferrucci and Balducci (2008) state that “aging is associated with increased incidence and prevalence of anemia” and, “in approximately 30% of cases, anemia in older individuals is due to either relative or absolute erythropoietin deficiency”.

Both the astronauts and the elderly tend to experience problems related to articular cartilage. The deleterious effects of microgravity on the human skeleton and more specifically on cartilage are known. “Data from terrestrial hind-limb unloading (HLU) animal experiments and human bed-rest studies demonstrate that reduced mechanical forces associated with joint unloading and immobilization leads to the proteoglycan loss in articular cartilage” (Fitzgerald 2017). Lotz and Loeser (2012) emphasize the effects of aging on articular cartilage homeostasis, mentioning extracellular matrix changes such as “reduced thickness of cartilage, proteolysis, advanced glycation and calcification” and cellular changes such as “reduced cell density, cellular senescence with abnormal secretory profiles, and impaired cellular defense mechanisms and anabolic responses”.

Finally, there are many studies on the weakening of sight, in both astronauts after long-term missions and the elderly. There is no need to provide literature confirming sight problems in seniors, as we all observe this phenomenon in everyday life. Several studies provide evidence that analogous problems may affect astronauts. By analyzing data concerning 300 individuals, Mader et al. (2011) show that “approximately 29% and 60% of astronauts on short- and long-duration missions, respectively, experienced a degradation in distant and near visual acuity” and add that “some of these vision changes remain unresolved years after flight”. The postflight inquiry documented that “after 6 months of space flight, 7 astronauts had ophthalmic findings, consisting of disc edema in 5, globe flattening in 5, choroidal folds in 5, cotton wool spots (CWS) in 3, nerve fiber layer thickening by OCT in 6, and decreased near vision in 6 astronauts.” The effect of microgravity on ocular structures and visual function has been reviewed in detail also by Taibbi et al. (2013).

After reading this literature, one wonders if the prevailing effect of spaceflight is aging or anti-aging. To put it in simple words, by flying for some time in space, does one become older or younger than one would become while remaining firmly on planet Earth? In terms of effects on the human body, microgravity is an environment that is still not completely understood. As we could see, the effects are mixed. Moving in a microgravity environment requires less effort, and this condition may benefit some organs. Besides, human cells and organs tend to adapt to the new condition, starting from gene expression (Wnorowski et al. 2019). Still, it seems that aging effects are largely prevailing. Even those articles that stress the anti-aging effects of

spaceflight recognize that all that glitters are not gold. For instance, if it is true that aging in *C. elegans* seems to slow down in space, it is also true that “male *Drosophila* lived shorter on the ground after spaceflight compared to controls that maintained on the ground through their lifetimes” (Honda et al. 2012).

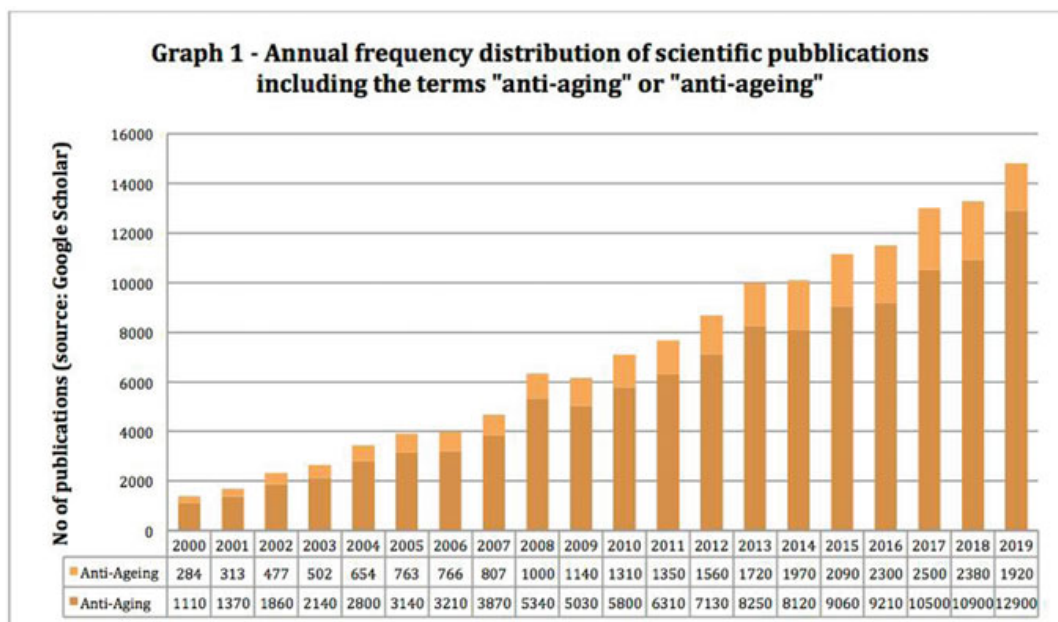
This does not mean, however, that humans should abandon their dreams of space exploration and colonization. There are at least three good reasons that justify a moderate optimism. The first reason is that we already possess quite effective sets of countermeasures to combat the negative effects of spaceflight (Iwamoto et al. 2005; Payne et al. 2007; Hargens et al. 2013). The second reason is that outer space is a “gold mine” and, consequently, private companies and governmental agencies will keep pursuing space exploration and colonization by using all the possible means offered by emerging technologies, including artificial intelligence and advanced robotics (Braddock et al. 2019; Campa et al. 2019). The third reason is that health problems related to spaceflight can also be countered by human enhancement and in particular artificial rejuvenation. Anti-aging medicine keeps progressing. And this is the main theme of this chapter. Our purpose is to show how breakthroughs in anti-aging medicine may shortly become a game changer for space exploration and colonization, helping to prevent or promptly recover from damages caused by hostile space conditions.

10.2 A Scientometric Glance on Anti-Aging Research

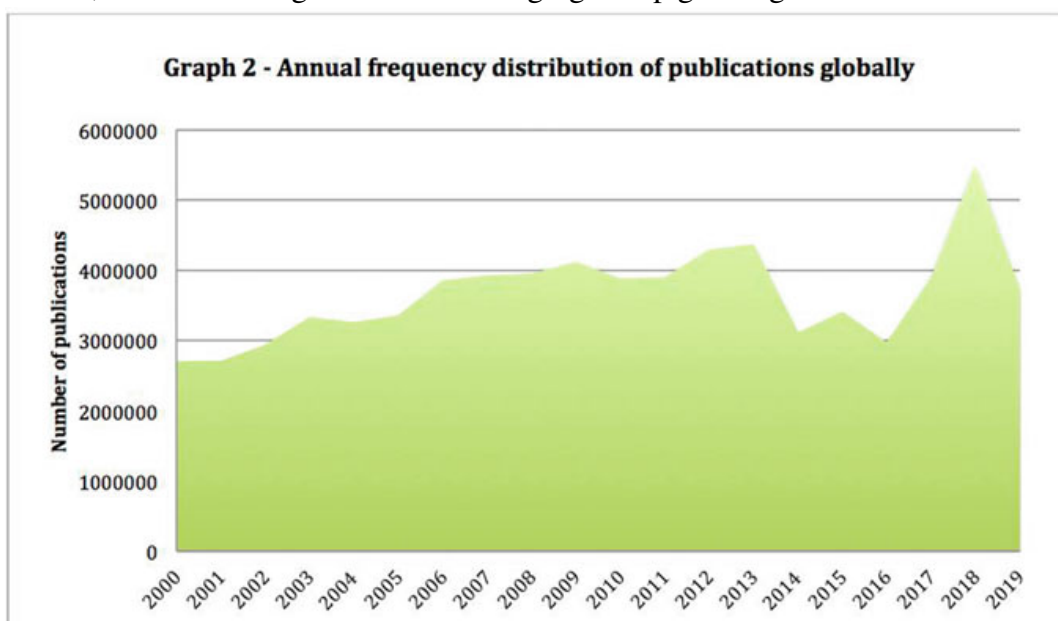
If one looks for the term “rejuvenation” in Google Scholar, the search engine finds around 300,000 results. It is clear however that this word can be used in very different contexts. A handmade search reveals that not a few of the detected scientific works focus on the rejuvenation of various activities and things, such as business management or libraries, rather than that of the human organism.

If the search is restricted to the combination of the terms “rejuvenation” and “medicine,” it still provides a large number of items—96,000—to be exact. Nonetheless, it seems safer to look for a more specialist term, such as “anti-aging”. In this case, Google Scholar detects 173,000 scientific works which include the word. If the search is limited to the twenty-first century, 76,500 books and articles on the subject are detected. Another 28,600 scientific works emerge, if we set the search on the British spelling of the word that is “anti-ageing”.

The most interesting aspect is the trend. As one can see from Graph 1, in the twenty-first century, the publications including the term “anti-aging” or “anti-ageing” experience a steady growth.

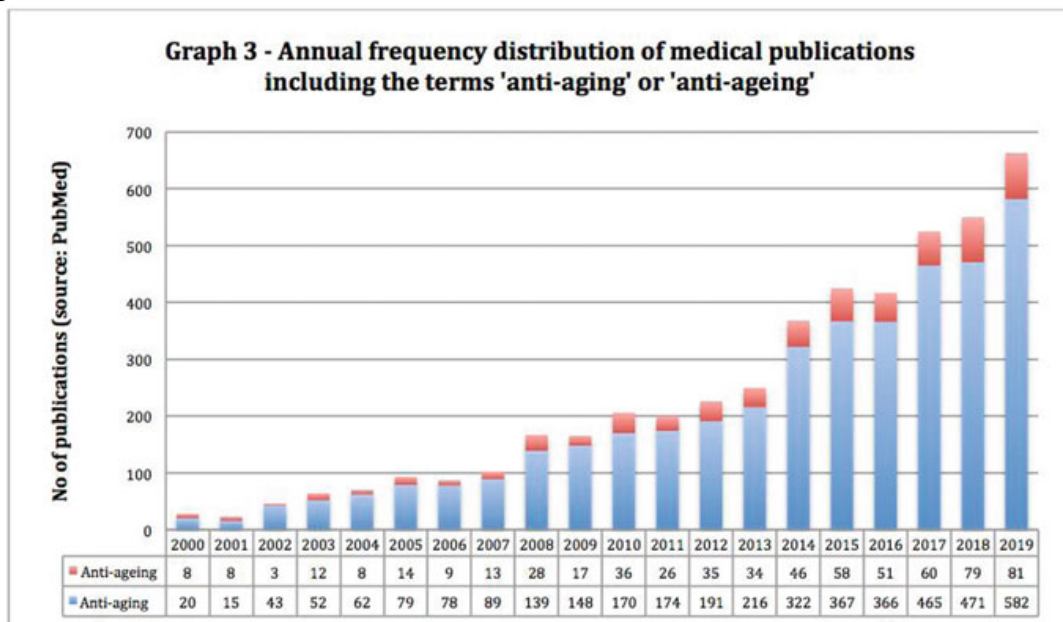


The figure represents the distribution of the absolute frequencies. Relative frequencies would tell us the proportion of the intellectual effort devoted to this research field. However, in this context, the global effort is more significant than the relative one. After all, it is the global effort that increases the likelihood of finding an effective solution for the aging problem, which is what matters. We can anyway exclude that the growth of publications in anti-aging medicine is a side effect of the general growth of publications. As Graph 2 clearly shows, the distributions of scientific publications globally follow a different pattern. Even when the absolute frequency of publications decreases, those including the word “anti-aging” keep growing.



This search could be implemented in different ways. For instance, we could include in the search the scientific literature in languages other than English, and the number of items would massively grow. Or we could restrict the search to the journals

with higher impact factors, and the numbers would fall. Google Scholar detects publications in any discipline. Indeed, anti-aging therapies can be discussed by physicians as well as philosophers or theologians. If we restrict the search to medical literature, the global number decreases. However, even if we rely on a more specialized search engine such as PubMed, which primarily accesses the MEDLINE database of references and abstracts on biomedical topics and life sciences, the number of articles remains remarkable. The medical publications including the term “anti-aging” are 3726 globally. Those published in the twenty-first century are 3617, while only 109 articles were published in the twentieth century. The search for the British word “anti-ageing” gives a similar result, even if on a smaller scale. The total amount of articles is 569, while those published in the twenty-first century are 548. The struggle against senescence has reportedly gained momentum in the new millennium. Graph 3 shows that the growth of medical publications is even more accentuated than the growth of all publications including the terms “anti-aging” or “anti-ageing.” If Graph 1 represents a steady growth, the curve in Graph 3 takes the shape of a seemingly exponential function.



If these are the numbers, it is understood that we cannot provide a complete literature review of the subject. Here, we will review just a few exemplary cases, to produce a picture of the most trendy ideas in the field of anti-aging medicine.

10.3 Regenerative Medicine 2.0

With regard to stopping or even reversing the senescence process, a first research line considered very promising is that of regenerative medicine. The basic idea is to regenerate organs and tissues by infusing reprogrammed stem cells in the human body. The fundamental breakthrough in this field dates back to the beginning of the

eighties when pluripotent stem cells are derived from the embryos of mice in two independent studies (Martin 1981; Evans and Kaufman 1981). After the discovery of the CRISPR/Cas-9 system, which allows us to reprogram the DNA as if it were software, we have entered a new phase of regenerative medicine (Li et al. 2018).

To take an example of regenerative medicine 2.0, in 2019, a group of researchers led by Ergin Beyret showed that CRISPR/Cas9 therapy can suppress aging, improve health, and prolong lifespan in mice. More precisely, researchers have developed a new gene therapy capable of suppressing the accelerated aging observed in mice with Hutchinson-Gilford progeria syndrome, a rare genetic disease that also affects humans (Beyret et al. 2019). The ultimate goal is to use this knowledge to halt the senescence process of the human organism.

Many independent research projects focus on the regeneration of single organs and tissues. Also, in 2019, the combined use of new genetic editing technologies and reprogrammed stem cell infusion has been successfully applied to the regeneration of skull bones (Truong et al. 2019).

As for the prospects in the near future, one thinks about the use of mesenchymal stem cells for tooth regeneration. A team of Chinese biologists remarked that “a number of basic researches, preclinical studies and clinical trials have investigated that dental stem cells efficiently improve formation of dental specialized structure and healing of periodontal diseases, suggesting a great feasibility and prospect of these approaches in translational medicine of dental regeneration” (Shuai et al. 2018).

Regenerating the entire organism, piece by piece, is the goal on which the efforts of thousands of researchers around the world converge. Advances in this field will become fully apparent when regenerative medicine is successfully applied to repair the visible parts of the body, especially teeth, hair, skin, and muscles. An octogenarian with the body’s internal and external organs completely regenerated, on a functional and aesthetic level, would de facto be a young person.

New skin regeneration techniques have been studied for years, to replace traditional plastic surgery. The book *Skin Tissue Engineering and Regenerative Medicine* by Albanna and Holmes (2016) IV presents the state of the art in this research field. Published in 2016, the volume does not include recent years’ innovations. In a rapidly growing sector like that of regenerative medicine, four years are significant amount of time. However, in this context, we are mainly interested in the basic idea. The authors start from the principle that the skin is the largest system of human organs and that “loss of skin integrity due to injury or illness results in a substantial physiologic imbalance and ultimately in severe disability or death”. Let us think about victims of burns or cut injuries. Tissue engineering and regenerative medicine are born with a therapeutic purpose. They are designed to deal with diseases, illnesses, and disabilities. However, it is not difficult to understand that these treatments, once perfected, would not remain confined to the clinical and therapeutic field. In principle, they can be used to reverse the senescence of epithelial tissues, to regenerate the skin and epidermis of elderly individuals, or to prevent aging in healthy individuals. The senescence process itself could eventually be seen as a disease.

10.4 The Frontier of Parabiosis

A second promising strand in the field of anti-aging therapies is that of heterochronic parabiosis. It is a field of study that has an ancient history, which had momentum in the second half of the twentieth century and then went out of fashion, to finally return to being explored again today. The reason for the discontinuity of this research topic could be found in the sinister aspects of the basic idea and of the experimental methods. To get straight to the point, if one injects the blood of a young individual into the body of an elderly one, the latter rejuvenates. This fact is proved by experiments on laboratory mice which undoubtedly look gruesome to animal rights activists. A certain portion of skin located on the lateral back of two rats, one young and one elderly, is connected through surgery. From that moment, the two animals form a single body. They must walk together, eat together, and sleep together. The capillaries of the skin are also connected so that the blood of the two animals is continuously mixed. After a period of symbiotic life, the rats are disconnected and the biological parameters re-measured. After the “treatment,” the elderly individual is significantly rejuvenated.

As a team of Italian researchers explain, “transfusion (or drinking) of blood or of its components has been thought as a rejuvenation method since ancient times. Parabiosis, the procedure of joining two animals so that they share each others blood circulation, has revitalized the concept of blood as a putative drug. Since 2005, a number of papers have reported the anti-ageing effect of heterochronic parabiosis, which is joining an aged mouse to a young partner” (Conese et al. 2017). The body’s ability to regenerate most tissues reaches its apical moment in youth. One of the reasons why we age is the gradual loss of this ability. The decline can be partially traced to the impairment of the function of stem and progenitor cells. Parabiosis experiments demonstrate “that factors derived from the young systemic environment are able to activate molecular signaling pathways in hepatic, muscle or neural stem cells of the old parabiont leading to increased tissue regeneration.” Going into technical details, the Italian researchers stress that “further studies have brought to identify some soluble factors in part responsible for these rejuvenating effects, including the chemokine CCL11, the growth differentiation factor 11, a member of the TGF- β superfamily, and oxytocin” (Conese et al. 2017). It should be, however, remarked that the mechanisms that produce the rejuvenation effects are still largely unknown. The explanation crafted by Loffredo, Wagers, and others, back in 2013, is under debate. For instance, it seems to be conflicting with the results published in *Cell Metabolism* by Eggerman, Glass, and others, in 2015. This controversy has been briefly summarized by Reardon on *Nature* (2015). Still, in spite of the fact that there is no unanimously accepted theory explaining the phenomenon, there is little doubt that heterochronic parabiosis empirically works quite well.

If it works with rats, it could or should work with humans too. Experiments can hardly be performed on humans in the same way as they are on rats. However, since the trials do not involve drugs, but transfusions, any private clinic can put the idea into practice, once it finds young volunteers willing to donate blood and

old patients wishing to have it injected. In the popular press, we already read of wealthy Silicon Valley managers who have no intention to wait for the conclusion of parabiosis clinical testing and are ready to buy young blood transfusions to reactivate the regeneration of their bodies (Dancyger 2018).

Conspiracy literature already speaks of “class vampirism”. We can certainly agree that a world in which everything is for sale, included blood, appears to be quite sinister, but we must not forget that research on artificial blood creation is quite advanced. In 2017, a group of Bristol scientists published an article in *Nature Communication* that shows how artificial blood can be produced in large quantities.

In the incipit of the article, Trakarnsanga et al. (2017) note that the lack of blood is an important global health problem that risks becoming even more problematic in the future because, in the face of the extension of people’s average life span, there is a decrease in the number of blood donors. That is why it is vital to find alternative methods to produce red blood cells.

Cultured red blood cells provide such an alternative and have potential advantages over donor blood, such as a reduced risk of infectious disease transmission, and as the cells are all nascent, the volume and number of transfusions administered to patients requiring regular transfusions (sickle cell disease, thalassemia myelodysplasia, certain cancers) could be reduced, ameliorating the consequences of organ damage from iron overload (Trakarnsanga et al. 2017).

As regards this challenge, many research teams are active and looking for the most effective and safe procedure. The mechanisms involved in the production of blood are still not fully understood, and different techniques under investigation have pro and contra. The British team recalls that “pluripotent stem cells (PSCs) provide a potentially unlimited progenitor source; however, there are substantial hurdles to overcome before these cells can be considered for manufacture of red cells”. One of the problems is “the small number of erythroid progenitors generated to date and severely impaired enucleation of the resultant erythroid cells”. The method presented in *Nature Communications* seems to overcome these problems. The strategy consists in generating immortalized adult erythroid progenitor cell lines. As the researchers emphasize, “such lines are capable of providing an unlimited supply of red cells and need only minimal culture to generate the final product”.

We will not discuss the details of the technique, as the matter goes beyond our scientific competence. What we mainly want to emphasize is that the idea of blood as a rejuvenation resource must be understood within the framework of an almost unlimited availability of artificial blood. Certainly, ethical problems will be raised also as regards this scenario. Such is the axiological dissonance within the Western society, especially in the bioethical field, that there is virtually no idea immune from controversy. To provide just an example, in a 2018 article published in the journal *Transfusion Medicine and Emotherapy*, Hofmann (2018) hastens to affirm that, “even if rejuvenating substances from blood may be artificially and cheaply produced and justly distributed, problems arise: survival may have to be balanced against reproduction, as reproductive age increases”. Hofmann admits that endless bliss and eternal youth have always been vital human dreams and that “parabiosis may bring us closer to the fountain of youth than ever,” but at the same time, he stresses

that “it is still too early to provide full-fledged assessments of its implications or to foresee how it will change health, aging, medicine, and society”. Therefore, he invites us to discuss the possible scenarios opened up by this research line.

These observations are relevant, but it is also true that a pragmatic approach gives priority to the solution of real current problems, rather than hypothetical future ones. If parabiosis understood as class vampirism rightly arouses a feeling of rejection, once it is possible to produce young blood in unlimited quantities in the laboratory, we can expect that the treatment will remain ethically problematic only for a marginal fringe of the population. After all, there is already a percentage of people who, for religious reasons, refuse blood transfusions even for therapeutic purposes.

10.5 Growth Hormone, or the Holy Grail

Growth hormone (GH), or somatotropin, has been known for more than half a century as a potential elixir of eternal youth. It is well known that this peptide produced by the pituitary gland stimulates statural growth during childhood and adolescence. Somatotropin continues its action during adulthood, favoring the oxidation of lipid stocks and regulating various metabolic processes. In simple words, it helps to keep the body strong, fit, healthy, and young. The problem is that the amount of hormone secreted by the gland decreases over time. During childhood, the secretion of somatotropin grows and, not surprisingly, during puberty reaches its maximum, to remain constant during the young age. After the thirtieth year of age, the body begins instead to reduce the secretion of growth hormone. In a fifty-year-old individual of the human species, the secretion is halved compared to twenty years before, while in a seventy-year-old individual, the secretion of somatotropin is reduced to a third if compared to that measured when he was a young adult. The decline in production is dramatic, with all the consequences that each of us can ascertain.

There is evidence that this process is not inevitable. Lifestyle greatly affects the secretion of growth hormone. For example, a rich protein diet and heavy physical exercise cause the pituitary gland to secrete a greater amount of somatotropin, counteracting the physiological decline and aging of the body. However, the increase in secretion is greater in untrained subjects than in trained ones, because it is activated in the presence of a considerable and painful effort, capable of stimulating the production of lactic acid. Put it differently, once a certain degree of physical fitness is reached, training does not produce appreciable rejuvenating effects. Therefore, alternative solutions to endogenous production have always been sought, such as the intake of anabolic substances capable of stimulating the secretion of the hormone, or the intravenous injection of the hormone itself. However, the latter solution also presents complications.

To show what the problems are, let us briefly review the history of the discovery. In the 1950s, in Montreal, John C. Beck and some collaborators carried out clinical studies on the growth hormone of a thirteen-year-old male specimen of the *Macaca mulatta* species, or rhesus macaque. At a time when experimental protocols and

examinations of ethical committees were less rigid and rigorous than those of today, the step from animal testing to human experimentation was relatively short. A first study, published in *Science* in 1957, showed that the administration of human and monkey growth hormone leads to a significant improvement in the conservation of nitrogen, to a retention of potassium, phosphorus, calcium, and sodium, to an increase in body weight and, ultimately, a significant increase in urinary aldosterone excretion. The effects were more pronounced when the human growth hormone was administered (Beck et al. 1957).

In 1958, on the *Annals of Internal Medicine*, the same research team published a similar study conducted on six people: an eighteen-year-old male with a statural and sexual delay and a skeletal age between thirteen and fourteen; a fifteen-year-old female with well-documented hypopituitarism secondary to craniopharyngioma; a fifty-three-year-old woman with breast cancer and widespread skeletal metastases; a sixty-eight-year-old woman with advanced postmenopausal osteoporosis; and a healthy twenty-four-year-old medical student. Once again, the ability of the substance to visibly affect metabolic processes was proven (Beck et al. 1958).

Since the first attempts to synthesize the growth hormone in the laboratory failed, the substance was initially extracted from the pituitary glands of corpses. The difficulty in finding the peptide limited its use for therapeutic purposes. It was mainly administered to patients suffering from idiopathic short stature. However, in 1985, it was discovered that subjects who had received the growth hormone extracted from corpses ten or fifteen years earlier often suffered unexpected pathologies. Unusual cases of Creutzfeldt–Jakob disease, better known as “mad cow disease,” were diagnosed. The cadaver-derived growth hormone was therefore removed from the market. The hypothesis is that the infectious prions that cause the disease can be transferred together with the somatotropin extracted from the gland.

This gruesome discovery, however, did not stop the use of the substance, because just in those years, the procedure to produce biosynthetic growth hormones was eventually devised, through the recombinant DNA technique. The discovery is credited to Choh Hao Li, a Chinese biochemist, emigrated to the USA in 1935. In the mid-1960s, the Chinese scientist discovered that somatotropin is composed of a chain of amino acids. In the following decade, Choh Hao Li managed to identify, purify, and synthesize the hormone (cf. Cole 1996).

The pharmaceutical industry immediately comes into action. The Californian company Genentech paves the way to the first use of recombinant human growth hormone for therapeutic purposes. In 1981, Robert A. Swanson, President and Chief Executive Officer of the company, announces that three products—human insulin, human growth hormone, and interferon—all produced by microbes developed at Genentech, undergo human clinical trials (Genentech 1981). A little later, Nutropin is put on the market.

One of the practical problems related to this treatment is the need to administer the substance through injections. Oral administration would be ineffective because the substance would simply be digested. Injecting it daily, however, requires a continuous rotation of the areas in which to insert the needle or the insertion of a subcutaneous cannula fixed with a patch that can be used several times. In any case, the treatment

can cause discomfort. Attempts to solve the problem fail. In 1999, the FDA approves Nutropin Depot, a drug patented by Genentech and Alkermes which provides for the prolonged release of the substance. This product requires an injection every two weeks, if not every month, instead of the daily treatment. However, Nutropin Depot is not successful, and in 2004, its sale is suspended. The production cost is too high and the price non-competitive, if compared to that of other products on the market.

Since 2005, many other recombinant DNA growth hormones have been synthesized. The list of approved somatropin-based products includes Pfizer's Genotropin, Eli Lilly's Humatrope, Novo Nordisk's Norditropin SimpleXx, Ipsen's NutropinAq, Merck Serono's Saizen, Ferring's Zomacton, and Sandoz Omnitrope. In addition to medical treatments, the potential use of GH for sports doping purposes was immediately glimpsed. The peptide hormone is now easier to find than when it was extracted from cadavers. Besides, the substance naturally produced by the adenohypophysis is difficult to distinguish from that surreptitiously introduced into the body. Although it remains an expensive treatment, the sport industry can sustain the expense and is constantly exposed to the temptation of doping, as the profitability of this business is mostly dependent on athletes' performance. As was the case with testosterone and anabolic steroids, the issue is constantly being examined by scientific literature and new strategies are continuously studied to avoid abuse (Holt 2011; Baumann 2012).

Abuses can also occur outside the sports world. Since it is known that this hormone has anti-aging effects, whoever has the necessary financial resources tries to obtain it from compliant doctors or from patients with medical prescription willing to resell it, with all the risks of the case, being the clinical tests still in progress.

The idea of pharmaceutical companies is to finally arrive at a safe product that can be legally sold also to healthy individuals. This path is leading to surprising results. On September 8, 2019, we learned that a group of Californian researchers has succeeded not only in stopping aging in adult human beings but even in reversing the natural process of senescence. The subjects of the experiment have been rejuvenated. The news quickly bounced from *Aging Cell*, the scientific journal in which it appeared, to traditional media and social networks.

In short, it seems possible to turn back the clock on biological age. The treatment had already been tested with encouraging results on laboratory animals, but it was still not known whether it was possible to reduce the biological age of adult human organisms in safe conditions. The study reports a clinical trial of Thymus Regeneration, Immunorestitution, and Insulin Mitigation (TRIIM), involving nine male individuals of Caucasian ethnicity, aged between fifty-one and sixty-five. The experiment was conceived and led by the immunologist Gregory Fahy, scientific director and co-founder of Intervene Immune, a private company operating in the field of biotechnology and health services based in the Los Angeles area. The trial took place at the Stanford Medical Center in Palo Alto, California.

For a year, nine healthy volunteers took a "cocktail" of three substances: growth hormone and two common diabetes drugs. At the end of the trial, the biological age of the patients was measured by determining the state of genome DNA methylation. On average, the volunteers resulted to be two and a half years younger. Their immune

system also benefited from the therapy, showing signs of rejuvenation (Fahy et al. 2019).

Of course, a certain degree of skepticism is always advisable. For decades, we have been hearing that we are on the threshold of the lost paradise, one step away from achieving eternal youth or unlimited life. News on the amazing results of revolutionary anti-aging therapies circulate all the time, even though, unfortunately, not a few of the treatments on the market are scams. As is well known, private companies are interested in patenting treatments, to profit from them. As a consequence, they sometimes emphasize positive results and minimize negative ones, to attract investments or get approvals from drug agencies.

Caution is a must even when faced with serious research, as the one published in *Aging Cell*. The Californian researchers warned that those achieved are only preliminary results, obtained by a trial on a small sample of volunteers and in the presence of a conflict of interests. Therefore, further studies on more representative samples, including subjects of different age, sex, and ethnic groups, are needed. Studies performed by independent researchers would also be needed.

Still, there are reasons for moderate optimism. The project was screened by ethics committees. The trial was approved, back in 2015, by the FDA. The study passed the double-blind assessment process and was published in a prestigious journal. Solid foundations are laid for further experimentation. The news was also taken up by *Nature*. The methods and results of the experiment were summarized as follows:

The latest trial was designed mainly to test whether growth hormone could be used safely in humans to restore tissue in the thymus gland. The gland, which is in the chest between the lungs and the breastbone, is crucial for efficient immune function. White blood cells are produced in bone marrow and then mature inside the thymus, where they become specialized T cells that help the body to fight infections and cancers. But the gland starts to shrink after puberty and increasingly becomes clogged with fat. (Abbott 2019).

Growth hormone stimulates thymus regeneration but in the presence of unwanted side effects, such as the contraction of diabetes. Therefore, “the trial included two widely used anti-diabetic drugs, dehydroepiandrosterone (DHEA) and metformin, in the treatment cocktail.” The results were a surprise for the researchers involved in the experimentation. Epigenetic analysis was conducted by geneticist Steve Horvath of the University of California, Los Angeles, who developed a multi-tissue predictor of age that allows us to estimate the age of DNA methylation of most tissues and cell types (Horvath 2013). His statement, also reported by *Nature*, ignites more than one hope on the effectiveness of this therapy: “I’d expected to see slowing down of the clock, but not a reversal”. And he concluded: “That felt kind of futuristic”.

10.6 Lengthening of Telomeres

In the scientific community, strong is the conviction that we get old because the cells that make up our body lose the ability to divide over time. One of the mechanisms

identified at the base of this phenomenon is the shortening of telomeres, which are protective caps at the ends of chromosomes. Telomeres shorten with each cell division, and the time comes when they are so short, they damage the same genes they were supposed to protect. At that point, the cell stops dividing.

This mechanism works at different speeds in different individuals, not only for genetic reasons but also as a consequence of lifestyle or psychological stress level. Hence, it is useful to distinguish between chronological and biological age of a person. Telomere length is considered a marker of biological age. People having longer telomeres have a longer life expectancy and run less risk of developing age-related illnesses, such as Alzheimer's and Parkinson's disease, or cancer. From these considerations, the idea was born that "eternal youth" can be achieved by finding a way to slow down or stop the shortening of telomeres, or even cause their elongation. There is evidence that telomeres can change length faster than previously thought. Lengthening can also be achieved through specific mental or physical training and can be observed after a few months of treatment. A team of researchers has indeed discovered that the change in telomere length is associated with structural changes in the brain. The experiment was performed on 298 healthy adults, stimulated through mental activities and constantly monitored. When telomeres lengthen, with a certain degree of probability, the cerebral cortex also thickens, while telomere shortening is associated with a reduction in gray matter. As the researchers claim, their study "provides the first evidence to date for an association between short-term change in leukocyte telomere length and brain structure, suggesting that these processes may be mechanistically linked; the mental training used did not influence leukocyte telomere length of healthy, middle-aged adults" (Puhlmann et al. [2019](#)).

The researchers also wanted to verify if the natural association between phenomena could be triggered by a mental training based on the growth of awareness and empathy, for a duration of nine months. As explained by the Max Planck Institute for Human Cognitive and Brain Sciences, "previous data from the ReSource Project, which was supported by the European Research Council (ERC), had already shown that certain regions of the cortex can be thickened by training, depending on the respective mental training contents of three distinct modules, each lasting for three months" (Max Planck [2019](#)).

However, the researchers admitted that the mental training used did not affect the length of leukocyte telomeres in healthy, middle-aged adults, and therefore, other studies are needed to fully understand the mechanism and eventually control it.

Anyway, mental training is just one of many different strategies to elongate telomeres. A pharmacological treatment is also on the agenda of biologists and physicians. For instance, several studies have shown that there is a correlation between lithium intake and longevity, even if the mechanisms underlying this phenomenon have not yet been perfectly understood. Lithium is naturally found, in different concentrations, in tap water and in the food we eat, particularly in vegetables and derived food. The concentration of lithium in drinking water varies in different regions of the planet. In 2011, a team of Japanese researchers used "weighted regression analysis to identify putative effects of tap water-derived lithium uptake on overall mortality" (Zarse et al. [2011](#)).

Independently, the same team of scientists has exposed some specimens of *C. elegans*, a small nematode commonly used for anti-aging studies, to comparable concentrations of lithium, quantifying their mortality. The results appeared encouraging. As for humans, “an inverse correlation between drinking water lithium concentrations and all-cause mortality in 18 neighboring Japanese municipalities with a total of 1,206,174 individuals ($\beta = -0.661, p = 0.003$)” was found. Consistently, it was also found that “exposure to a comparably low concentration of lithium chloride extends life span of *C. elegans* ($p = 0.047$)”.

In the wake of this and other similar studies, the idea was born to verify if there is a correlation between lithium intake and telomere lengthening. A team of researchers led by Coutts (2019) made it clear that telomere length is both “a promising biomarker for age-related disease and a potential anti-ageing drug target,” and provided evidence that lithium may “confer its anti-ageing effects by moderating the expression of genes responsible for normal telomere length regulation”.

As regards the theoretical basis of the study, it is worth recalling that gene-enrichment analysis has identified thirteen genes associated with telomere length. Furthermore, the association between lithium and longevity can be measured not only on healthy subjects taking moderate amounts of lithium—through tap water, fruit, or vegetables—but also on subjects exposed to significant amounts of this substance, such as individuals subject to bipolar disorder. The depression that occurs during bipolar disorder is often treated with lithium. In this type of patient, antidepressants can cause manic episodes, while the response to lithium-based treatments is in most cases positive. The researchers confirmed that, concerning lithium, “chronic use in a sample of 384 bipolar disorder patients is associated with longer telomeres ($p = 0.03$)”.

To corroborate the theory that lithium in our tap water has beneficial effects on health and longevity, and that lithium’s effect on telomere length may be one of the mechanisms by which it confers its anti-aging properties, the researchers “tested whether lithium affects the expression of genes responsible for telomere length maintenance”. These were identified, from their gene-enrichment analyses, “in a relevant model system that recapitulates the drug’s anti-ageing effects”. Coutts and her collaborators found “that 3 out of the 13 genes identified from the gene-enrichment analysis had an assayed ortholog in a *C. elegans* model of lithium-induced extended longevity” and that “lithium had an effect on all three genes”.

Other studies show how lithium exposure can have anti-aging effects and reduce mortality in distinct species, by acting on the length of telomeres, but the examples just provided may suffice for our purposes.

The question that arises is whether one can elongate telomeres by directly modifying the genetic makeup of individuals, employing the new gene-editing techniques. It has been proved that the length of telomeres is also linked to hereditary characteristics, and not only to lifestyle or nutrition. For example, it is known that children’s organisms of different sex or ethnicity show different behaviors in this respect (Ly et al. 2019). Apart from all the ethical questions that can be raised regarding a modification of the human species with the means offered by genetic engineering, it has also been asked if the game is worth the candle. As we said, the telomeres gradually

shorten to the point that the cell is no longer able to divide, but, at the same time, the cells accumulate mutations and anomalies over time. If genetically modified telomeres were created that do not shorten, the cells could multiply to infinity even when they are mutated or diseased. In other words, we may not grow old but may be more prone to tumors or other diseases. The “expiration date” that hangs over all of us could serve to avoid this scenario.

It is worth noting that the scientific literature has already given answers to this possible objection. A recent study in rats published in *Nature Communications* shows that an over-extension of telomeres does not lead to any undesirable side effects. Muñoz-Lorente et al. (2019) explain their experiment as follows:

We generated mouse embryonic (ES) cells with longer telomeres than normal (hyper-long telomeres) in the absence of genetic manipulations, which contributed to all mouse tissues. To address whether hyper-long telomeres have deleterious effects, we generated mice in which 100% of their cells are derived from hyper-long telomere ES cells.

The results show no ambiguity whatsoever. It has been observed that “these mice have longer telomeres and less DNA damage with aging”. Mice with hyper-long telomeres are slim and have low cholesterol and LDL levels, as well as better glucose and insulin tolerance. Besides, they also have “less incidence of cancer and an increased longevity”. Otherwise stated, the study reports that having telomeres much longer than normal, in addition to slowing aging, not only does not increase the probability of contracting tumors but also, on the contrary, makes the onset of the disease less likely, at least in mice.

10.7 Concluding Remarks

In light of the medical studies carried out on astronauts during and after space missions, ethical concerns and doubts were rightly raised regarding plans of long-term space travel and colonization of other celestial bodies, starting with the Moon and Mars (Szocik et al. 2019b). Is it legitimate to send astronauts to space for long periods when the chance that their health condition could be hopelessly compromised is high? The issue of security is fundamental. We often forget, however, that medical research progresses very quickly, and as we believe we have adequately demonstrated, anti-aging medicine seems to be on the verge of a scientific revolution. If it were possible to regenerate and rejuvenate an entire human organism, many doubts about the safety of long-lasting space missions would *ipso facto* vanish. What we have presented here, both from a quantitative and qualitative point of view, are just some of the research lines in the field of anti-aging medicine. Our goal was to clarify that when we speak of “eternal youth,” we are not talking about the regurgitation of the alchemical dream of a few visionaries, but of a huge scientific and industrial machine that has been set in motion in recent decades and is unceasingly producing new studies and discoveries. This is not to deny that there are still technical and ethical obstacles on the way. These scientific results could be questioned by new

studies, and ethical or legal limitations to this field of inquiry could be imposed by governmental agencies in the future (Campa 2019, 160–173). However, there are good reasons to believe that this massive deployment of public and private economic resources and intellectual efforts will lead to important advances in these fields.

As we said, some scholars think that the very idea of rejuvenation, and of human enhancement in general, is morally wrong. They object that an expiration date of the human organism is necessary to maintain a social and natural equilibrium. It is our opinion that this objection can be rejected not only on scientific grounds but also on philosophical ones. It is evident that, for the maintenance and natural evolution of the species, the individuals serve as long as they are fertile and can take care of the offspring. Then, they become useless. If we were to decide to go along with nature, assuming it to be wiser than human rationality, we would have to question the whole of medical science and the very idea of health and social security services. It is not clear, indeed, why antibiotics and surgery are acceptable, not to say retirement pensions and health insurances, while hormones and telomeres should never be touched.

Since humans first began using tools, our evolution ceased to be purely natural and became self-directed, although not always knowingly. In a sense, our species is artificial by nature (Campa 2013). And if this is so, it is advisable to keep the pragmatic principle firm. First, we solve the ascertained problems, and then, we discuss the hypothetical ones. There is no doubt that aging, aging-related diseases, and death are problems. For those who accept this pragmatic perspective, the priority is to find effective anti-aging therapies. Then, if other health pathologies or social problems should arise in the lifetime gained, every effort will be made to find a remedy.

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